



BCJ LEARN

Architecture & Operations Handover

How the platform is built, where it runs, and how to operate, maintain, and support it after handover.

Prepared for	BCJ Building Services - IT & Leadership
Prepared by	Ten80ten
Document	Operations Handover (Deliverable #10)
Version	1.0
Date	June 22, 2026

Contents

- 1** System overview
- 2** Architecture at a glance
- 3** The services that run the platform
- 4** Hosting & deployments
- 5** Database & data
- 6** Environment variables
- 7** Authentication & security
- 8** AI question generation
- 9** Email delivery
- 10** Scheduled jobs
- 11** File storage
- 12** Routine operations (runbook)
- 13** Monitoring & troubleshooting
- 14** Backups & disaster recovery
- 15** Support & escalation

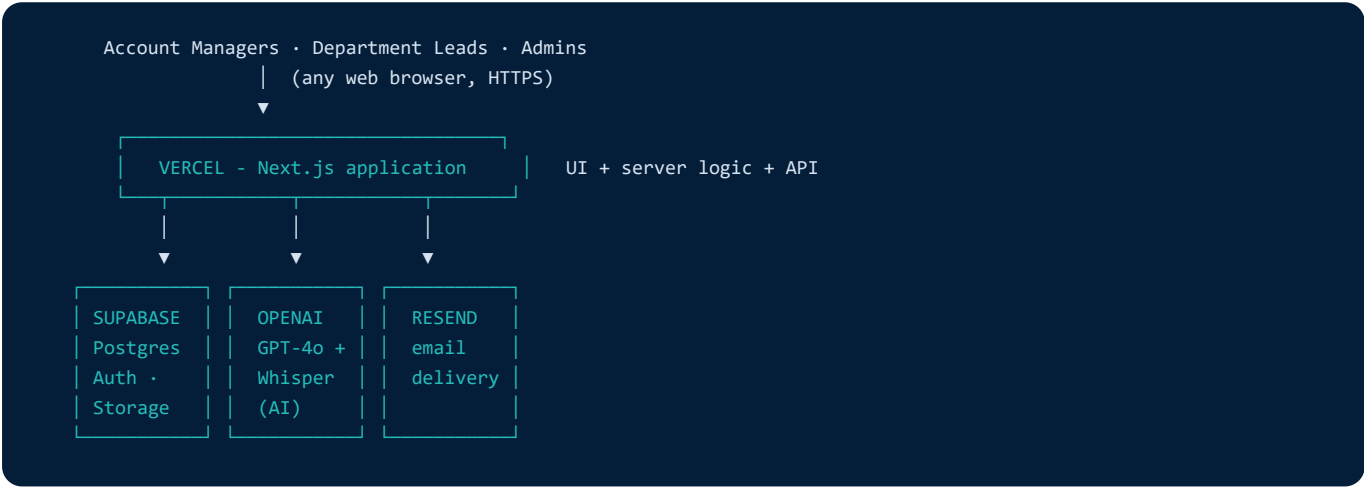
1 System overview

BCJ Learn is a custom, BCJ-owned web application for delivering the five-module Account Manager training programme: hosting content, running on-site quizzes, enforcing the 85% pass rule and retakes, and giving leadership live reporting. It is a single web app - no software for users to install - that runs on professional cloud services, all set up under BCJ's own accounts.

This document is for whoever maintains the platform on BCJ's side (IT, or a future support partner). It explains what the platform is built on, where each piece lives, and how to perform the common operational tasks. It does not require day-to-day code changes - the platform is built to run Modules 2-5 with no engineering work.

2 Architecture at a glance

The application is a **Next.js** web app hosted on **Vercel**. It talks to **Supabase** for its database, user accounts and file storage, to **OpenAI** for AI question drafting, and to **Resend** for email. Everything below is owned by BCJ.



LAYER	TECHNOLOGY	ROLE
Application	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS	The whole web app - pages, server actions, API routes.
Hosting	Vercel	Builds and runs the app; auto-deploys from GitHub; HTTPS, scaling, logs.
Database / Auth / Storage	Supabase (PostgreSQL)	All data, user accounts, sign-in, and uploaded files.
AI	OpenAI - GPT-4o & Whisper	Drafts quiz questions from content; transcribes videos.
Email	Resend	Sends invitations, sign-in codes, results and reminders.
Source code	GitHub (private, BCJ-IT)	The codebase; every change is versioned here.

3 The services that run the platform

Five external services power BCJ Learn, all under BCJ's own accounts. Login details are in the separate **Service Account Credentials** sheet.

SERVICE	WHAT IT DOES	SIGN IN AT
GitHub	Stores the source code (account: BCJ-IT)	github.com
Vercel	Hosts & deploys the live site	vercel.com
Supabase	Database, user accounts & file storage	supabase.com
Resend	Sends all platform emails	resend.com
OpenAI	AI question drafting & video transcription	platform.openai.com

4 Hosting & deployments

The app is hosted on Vercel and connected to the GitHub repository. Deployment is automatic:

- A change is pushed to the GitHub repo's main branch.
- Vercel automatically builds it and, if the build passes, publishes it to the live URL within a couple of minutes.
- Every deployment is kept in history, so a previous version can be restored instantly.

To roll back: Vercel dashboard → *Deployments* → pick a known-good deployment → *Promote to Production*. No code or rebuild needed.

No manual server administration is required. Vercel handles HTTPS certificates, scaling and uptime automatically.

5 Database & data

All data lives in a single **Supabase PostgreSQL** database - user accounts, modules, content, questions, quiz attempts, results, resources and notifications. Key points for whoever maintains it:

- **Schema changes** are managed as numbered SQL migration files in the codebase under `supabase/migrations/` (52 to date). They are applied in order via the Supabase SQL editor.
- **Row-Level Security (RLS)** is enabled on the tables: the database itself enforces who can read or write what (e.g. an employee can only see their own attempts). This is a core security layer, not just app logic.
- **Business rules** such as starting a quiz and grading run as secure database functions (RPCs), so the rules hold even if called directly.
- **Quiz integrity:** the same question pool is served in a randomised order; the 85% threshold, the 3-attempt limit and the easier retake pool are all enforced server-side.

Care with the database: editing rows directly in Supabase bypasses the app's safeguards. Prefer the in-app Admin tools; use direct SQL only for maintenance, and take a backup first.

6 Environment variables

The app is configured through environment variables held in **Vercel → Project → Settings → Environment Variables** (never in the code). The actual values are secret and are not printed here.

VARIABLE	PURPOSE
NEXT_PUBLIC_SUPABASE_URL	Address of the Supabase project.
NEXT_PUBLIC_SUPABASE_ANON_KEY	Public client key - restricted by Row-Level Security.
SUPABASE_SERVICE_ROLE_KEY	Server-only admin key (bypasses RLS). Keep secret.
OPENAI_API_KEY	OpenAI key for AI question generation & transcription. Keep secret.
RESEND_API_KEY	Resend key for sending email. Keep secret.
RESEND_FROM_EMAIL	The "from" address all emails are sent from.
NEXT_PUBLIC_APP_URL	The site's base URL, used to build links inside emails.
NEXT_PUBLIC_DEMO_MODE	When true, skips sending real emails (for testing). Set false in production.
CRON_SECRET	Password that protects the automatic reminders job.

After changing any variable in Vercel, trigger a redeploy so the new value takes effect.

7 Authentication & security

- **Sign-in:** email + password, handled by Supabase Auth. Optional **two-step verification** sends a 6-digit code by email.
- **Password reset:** self-service from the login screen via an emailed link.
- **Roles:** Admin, Department Lead, Employee - enforced both in the app and at the database (RLS).
- **Encryption:** all traffic is HTTPS; data is encrypted at rest by Supabase.
- **The service-role key** is used only by trusted server code, never sent to the browser. Treat it, and all secret keys, like passwords.

8 AI question generation

Quiz questions are drafted by **OpenAI: GPT-4o** reads the uploaded module content and writes the question bank; **Whisper** transcribes uploaded videos so their content can be used too. Nothing is published without a human approving it in the app.

- The feature runs on the key in `OPENAI_API_KEY`. To run it on BCJ's own account, create a key at `platform.openai.com`, add it in Vercel, and redeploy.
- Cost is usage-based and billed directly by OpenAI to the account that owns the key - typically modest at this scale.

An Anthropic (Claude) integration also exists in the code but is not currently used; AI runs on OpenAI.

9 Email delivery

All automated email (invitations, sign-in codes, pass/fail results, reminders, at-risk alerts) is sent through **Resend**. The text of every email is editable by an Admin inside the app under *Notifications*.

- Emails are sent from the address in `RESEND_FROM_EMAIL`.
- For best deliverability the sending domain should be verified in Resend (DNS records added to BCJ's domain). Until then, sending may be limited.

10 Scheduled jobs

One automated job runs on a schedule via **Vercel Cron**:

JOB	ENDPOINT	SCHEDULE
Overdue / reminder emails	/api/cron/reminders	Weekdays, 13:00 UTC

The endpoint is protected by `CRON_SECRET` so only Vercel can trigger it. The schedule is defined in `vercel.json`.

11 File storage

Uploaded files are kept in **Supabase Storage**, organised into three buckets:

BUCKET	HOLDS
module-content	Module manuals, slides, documents and videos.
branding	The platform logo and brand assets.
avatars	User profile photos.

12 Routine operations (runbook)

TASK	HOW
Add an Admin / Lead / Employee	In the app: <i>People</i> → invite. No code needed.
Publish a new module	In the app: build content, generate & approve questions, choose attendees, Publish.
Deploy a code change	Push to the GitHub repo's main branch → Vercel builds & deploys automatically.
Roll back a release	Vercel → Deployments → promote a previous deployment to Production.
Rotate / update a key	Create the new key in the provider → update it in Vercel → Environment Variables → redeploy.
Apply a database change	Run the relevant supabase/migrations/ file in the Supabase SQL editor.
Restore data	Supabase → Database → Backups → restore a point in time.

13 Monitoring & troubleshooting

- **App errors / logs:** Vercel dashboard → the latest Deployment → *Logs* (runtime & function logs).
- **Database & auth logs:** Supabase dashboard → *Logs*.
- **Email delivery:** Resend dashboard → *Emails* (see what sent, bounced or failed).
- **In-app activity:** the Admin *Audit log* records sign-ins, passes, publishes, reminders and more.

Common first checks: a failed AI generation usually means the OpenAI key is missing, unfunded or invalid; emails not arriving usually means the sending domain isn't verified or demo mode is on.

14 Backups & disaster recovery

- **Database:** Supabase takes automated backups (retention depends on the Supabase plan). Restores are done from the Supabase dashboard.
- **Code:** the full history is in GitHub; any prior version can be restored or redeployed.
- **Deployments:** Vercel keeps every previous build for instant rollback.

Recommended: confirm the Supabase plan's backup retention meets BCJ's needs, and keep at least one Admin and the IT account credentials stored in a company password manager.

15 Support & escalation

During the **30-day warranty period** after launch, Ten80ten fixes any defect - anything not behaving as described in the project scope - at no cost. New features follow the agreed change process and are quoted separately.

NEED	CONTACT
Day-to-day technical issue / bug	Furqan (Ten80ten - Technical Lead)
Commercial / scope / support tier	Alex Nicholson (Ten80ten) · +1 239-784-5377

After the warranty period, BCJ can self-manage the platform or arrange an ongoing support arrangement with Ten80ten, as set out in the project scope.

BCJ Learn - Architecture & Operations Handover · v1.0 · Confidential - for BCJ Building Services internal use only. Prepared by Ten80ten, June 22, 2026.